

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Renewable energy technologies - Set 1

Mark scheme

AQA spec 3.3.4.1; Energy resources; textbook pp. 196-217

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Photons of light absorbed by the (n-type silicon) upper layer dislodge electrons, making it negatively charged • Electrons flow through an external conductor / circuit to the relatively positive lower layer; the current powers appliances	2
01.2	• Smooth cells reflect about 30% of the light; grooved / textured surfaces (mimicking moth eyes) reflect light into the cell rather than away • Multi-junction cells have several layers of different materials, each absorbing different wavelengths • So a greater proportion of the available light is absorbed and converted to electricity	3
01.3	• Solar farm: polycrystalline / thin film - lowest cost per unit electricity where land area is not limiting (efficiency per m² matters less) • Satellite: multi-junction - highest efficiency where mass / area is critical and cost is less important • Cost per watt = cost / efficiency: thin film 0.125, poly 0.125, mono 0.136, multi-junction 0.625 per unit	3
01.4	• Intermittent: no output at night and low in winter (daily and seasonal cycles) - predictable • Unpredictable: cloud cover, dust and smoke reduce intensity and scatter light and cannot be forecast accurately	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1.	
02.1	• A fluid in a buried ground loop absorbs low-temperature heat from the ground; at the expansion valve the fluid boils at low pressure absorbing this heat • A compressor raises the pressure so the gas condenses, releasing heat at a higher temperature inside the building • The electricity only moves heat rather than creating it, so up to four times as much heat is delivered as electricity used (less if the source is colder)	3

Question	Marking guidance	Marks
02.2	- Parabolic reflectors focus sunlight onto tubes of oil, increasing the energy density and reaching high temperatures - The heat is transferred to molten salt (potassium nitrate) stored in large insulated tanks at up to 550 °C - The stored heat boils water to drive steam turbines whenever electricity is required, including at night	3
02.3	- Reflectors can only focus parallel rays of direct sunlight; cloud, dust and smoke scatter light so it cannot be concentrated - PV cells convert diffuse as well as direct light, though output is lower	2
02.4	Any 2 from: - Manufacture: extraction / processing of silicon, metals and plastics; toxic wastes (silicon tetrachloride, cadmium) - Land take by large solar farms competing with other uses / affecting desert flora and fauna - Water for cleaning panels in arid areas - Disposal of old panels	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 200-202; AQA spec 3.3.4.1.	
03.1	- No large dam / reservoir needed so no flooding of farmland, homes or habitats and no barrier across the whole river (run-of-river / micro-hydro diverts part of the flow) - Suitable for rivers with a small drop in height and lower flow rates that are common in lowland rural areas - Small scale matches local demand; electricity used locally without long transmission lines / grid connection - Low construction cost and simple technology maintained by the community - Helical turbines tolerate high turbidity / debris without damage and fish pass through unharmed; Kaplan turbines with adjustable blades cope with variable flow	5
03.2	Lack of capital / finance, technical knowledge or grid access; seasonal drying of rivers; conflicts with irrigation water; planning / land ownership	1
03.3	- Output is proportional to flow x efficiency: $100 \times 0.10 \times (40 / 90)$ = 4.4 kW (accept 4.4-4.5)	2
03.4	Any 2 from: - Constant regulated flow removes seasonal extremes: dry-season sandbanks used by nesting birds / turtles lost; high flows that clear gravel spawning beds lost; floodplain no longer flooded - Sudden flow increases for peak generation raise turbidity, wash away species, reduce light for photosynthesis - Sediment trapped so less nutrient replenishment downstream; barrier to migrating fish (salmon, sturgeon) and isolation of populations	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 203-205; AQA spec 3.3.4.1.	
04.1	$10^3 / 6^3 = 1000 / 216$ = 4.6 times (accept 4.63)	2

Question	Marking guidance	Marks
04.2	• C has the highest velocity (most energy, and energy rises with the cube of velocity), strong reliable winds over water, few land-use conflicts, but long expensive cable and offshore installation / maintenance costs • A is windy but on a migration route (bird strikes) and peat bog (foundations / tracks damage hydrology; carbon release); far from grid; scenic objections • B is close to the grid with good wind but near a village (noise, visual objection, shadow flicker) - possible with careful siting • D is cheap to connect and land can still be farmed between turbines, but low velocity gives little energy • Judgement, eg C or B; turbines spaced 3-5 blade diameters apart to avoid wind shadow	5
04.3	<i>Any 2 from:</i> • Driven by wind from any direction so no motor to turn them; suitable for turbulent urban sites • Generator can be at ground level so a lighter tower • Quieter • Operate at lower wind velocities • Blades more visible so fewer bird deaths	2
04.4	• No gearbox: fewer breakdowns, less friction so start generating at lower wind speeds, quieter, lighter nacelle (DC converted to AC by an inverter)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 206-209; AQA spec 3.3.4.1.	
05.1	• High mean wind velocities that are consistent in strength and direction • A long fetch (distance of open water) over which waves build up, eg Atlantic coasts of the UK	2
05.2	• Oscillating water column (eg Islay Limpet): waves force water up and down in a submerged chamber; the air forced in and out drives a turbine • Or point absorber (floating buoy moving relative to a fixed base turns a generator); surface attenuator (Pelamis - hinged sections push pistons pumping fluid over a turbine); overtopping device (Wave Dragon - waves fill a reservoir above sea level which drains through turbines); oscillating wave surge converter (Oyster)	2
05.3	<i>Any 2 from:</i> • Equipment must withstand storms and corrosion • Difficult to anchor off stormy coasts / in deep water • Expensive to transmit electricity from isolated coasts to consumers • Competition with established technologies	2
05.4	• Barrage: huge quantities of construction materials; reduces tidal range so intertidal mudflats used by wading birds are lost; changes sedimentation (erosion near turbines, deposition elsewhere), lower turbidity; pollutants concentrate in the lagoon; blocks migratory fish and shipping • Lagoon: similar but smaller scale; only part of the estuary enclosed so the most sensitive areas and main channel can be avoided • In-stream turbines: do not block the tidal flow so little impact on range or sediment; but noise affects cetaceans and many dispersed turbines could have a larger total impact; harness less energy	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 210-211, 214-217; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
06.1	- When there is a difference in level between the sea and the basin: on the ebb (about 6-8 hours) water flows out through the turbines - Also possible on the flood when the sea is higher than the basin; no generation when levels are equal (slack water)	2
06.2	- Two periods of slack water in each 12.5-hour cycle (twice in about 25 hours) give no flow; the timing does not match demand - Tides depend on the positions of the Moon and Sun which are known, so output can be forecast years ahead and alternative supplies planned	2
06.3	- Spring tides: Sun and Moon aligned so their gravitational effects combine giving a large tidal range; neap: effects oppose giving a small range - Energy depends on the mass of water and height difference, so output is much greater at springs	2
06.4	- When demand is low, surplus electricity pumps water into the lagoon above sea level - The stored water is released through the turbines later when demand is high (pumped storage)	2
06.5	- Large tidal range (eg Severn Estuary) - Estuaries, bays or headlands / channels that focus the flow and increase velocity	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 214-215; AQA spec 3.3.4.1.	
07.1	<i>Any 3 from:</i> - Supply rate of energy crops can be controlled and fuels stored so supply matches demand (not intermittent) - Liquid biofuels (alcohol, biodiesel) can replace petrol / diesel in vehicles - Carbon neutral in principle: CO₂ released equals CO₂ absorbed in photosynthesis - Energy density of alcohol / biodiesel nearly as high as fossil fuels; wastes (straw, sewage, landfill gas) are cheap	3
07.2	<i>Any 3 from:</i> - Large areas of farmland compete with food production or cause clearance of wildlife habitats (eg palm oil) - Intensive farming inputs (fertiliser, pesticide, machinery fuel) may release nearly as much CO₂ as using fossil fuels directly - Low energy density of straw, wood, Miscanthus makes transport uneconomic - must be used locally - Supply from wastes limited to the amount produced	3
07.3	<i>Any 2 from:</i> - Hydrogen from algae: some algae deprived of sulfur produce hydrogen during photosynthesis - Anaerobic digestion extended to crop wastes and marine algae; small-scale digesters (eg one cow's manure supplying a family's cooking gas) - Biofuels from microorganisms (eg oil-producing microalgae)	2
07.4	- Conventional geothermal steam turbines need rocks above 150 °C (recent volcanic activity), too deep to reach economically in the UK - New turbines use fluids such as butane / pentane that boil at low temperatures, so water as cool as 60 °C from deep boreholes / hot dry rock systems can drive them	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 212-214; AQA spec 3.3.4.1.	
08.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Properties of renewables: intermittency, predictability, low energy density, storage, applicability, locational constraints, impacts, technology level, economics • Wind: large offshore resource in shallow seas, falling costs, HAWT / VAWT / direct drive developments; intermittent, needs storage and grid; land-use conflicts • Solar: PV costs falling rapidly, roof space, but low winter insolation in the UK; CSP unsuitable; heat pumps and photothermal for heating • HEP: most large sites used; low-head, helical and micro-hydro extend use; pumped storage vital • Tidal: large range (Severn, Mersey), predictable; barrages costly with high impact, lagoons and in-stream turbines lower impact • Wave: long Atlantic fetch, but technology immature • Biofuels: wastes, anaerobic digestion, energy crops vs food / habitat • Geothermal: limited to low-temperature schemes / heat pumps • Systems issues: storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management, diversity of sources, energy conservation reducing demand, electrification of transport • Judgement: renewables could supply most electricity with storage and a diverse mix, but heat and transport are harder; transition needs investment and lifestyle change	9
08.2	• Eight times (energy proportional to velocity cubed)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-217; AQA spec 3.3.4.1.	